
Corrigé — Exercice 3

1) $\det(A) = (-1)(3) - (4)(2) = -11 \neq 0$, donc A est inversible et

$$A^{-1} = \frac{1}{-11} \begin{pmatrix} 3 & -4 \\ -2 & -1 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} -3 & 4 \\ 2 & 1 \end{pmatrix}.$$

2) $X = A^{-1}B = \frac{1}{11} \begin{pmatrix} -3 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 10 \\ 13 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} 22 \\ 33 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$

$$S = \{(2, 3)\}$$

3) $X = A^{-1} \begin{pmatrix} 2 \\ 7 \end{pmatrix} = \frac{1}{-11} \begin{pmatrix} 3 & -4 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 7 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} : (x, y) = (2; 1).$

4) $\det A = -11; x = \frac{1}{-11} \begin{vmatrix} 2 & 4 \\ 7 & 3 \end{vmatrix} = \frac{6 - 28}{-11} = 2$ et $y = \frac{1}{-11} \begin{vmatrix} -1 & 2 \\ 2 & 7 \end{vmatrix} = \frac{-11}{-11} = 1$: on retrouve $(2; 1).$